Absolute Reading Group 21.05

Nadja: the Artin-Schreier theorem

Thm (Artin-Schreier, 1927)

Let Ω be an algebraically closed field, let P be a proper subfield of Ω such that Ω/P is finite.

Then P is real closed & $\Omega=P(i)$ $i=\sqrt{-1}$

def

a field is real <-> -1 is not a sum of squares

real closed <-> real & no prop. alg. ext. is real

Kummer extensions & Artin-Schreier extensions: for p prime, these are cyclic extensions of deg p, either in characteristic p (Artin-Schreier) or in char $\neq p$ for fields containing primitive p^{th}-root of unity.

Artin-Schreier polynom: $X^p - X =: \wp(X)$

AS-ext & Kummer-ext are building blocks of finite ext.

Picture if $p \mid [K:k]$, $\exists F$: $k - F \overset{p}{-} K$

<u>Claim</u>: if $\mathrm{char}(K)=p$, any cyclic ext F/K of deg p is an AS-ext, i.e., $F=K(a)$ where $a^p-a-b=0$, $b\in K$.

& vice versa.

<u>Proof</u>: Let ϑ be a root of X^p-X-a, $a\in K$. Then $\vartheta, \vartheta+1 \dots \vartheta+(p-1)$ are all the roots of X^p-X-a.

So $K(\vartheta)$ contains all roots of $\wp(X)-a$.

if $\vartheta\notin K$, then $\wp(X)-a$ is irr. over K, so $K(\vartheta)/K$ is Galois of deg p & thus cyclic.

Now let $K(\alpha)$ be a cyclic extension of deg p over K.

Let σ be a generator of $\mathrm{Gal}(K(\alpha)/K)$, i.e., $\mathrm{Gal}(K(\alpha)/K)=\{\sigma^i\}$

Let $\alpha_i=\sigma^i(\alpha)$. $A=\begin{pmatrix} 1 & \alpha_0 & \alpha_0^2 & \cdots & \alpha_0^{p-1} \\ 1 & \alpha_1 & \alpha_1^2 & \cdots & \alpha_1^{p-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \alpha_{p-1} & \alpha_{p-1}^2 & \cdots & \alpha_{p-1}^{p-1} \end{pmatrix}$ Vandermonde Matrix

$\det(A)\neq 0 \iff \alpha_i\neq\alpha_j \implies \exists$ column which doesn't sum to 0: $s=\sum_{i=0}^{p-1}\alpha_i^k\neq 0$ for some k.

Let $\xi=-\frac{1}{s}\sum_{i=0}^{p-1} i\alpha_i^k$. <u>Claim</u>: $K(\xi)=K(\alpha)$ & ξ is an AS-root.

- $\xi \notin K$, because $\sigma(\xi) = -\frac{1}{s}\sum_{i=0}^{p-1} i\alpha_{i+1}^{k}$

(note that $\sigma(s) = s$) $\quad = -\frac{1}{s}\left(\sum_{i=0}^{p-1}(i+1)\alpha_{i+1}^{k} - s\right)$

$= \xi + 1 \neq \xi$

So $\xi \notin \mathrm{Fix}(\sigma) = K$!

therefore $K(\xi) \supsetneq K$ & $K(\xi) \subseteq K(a)$,

so $[K(\xi):K] \mid [K(a):K] = p$, so $K(\xi) = K(a)$.

- $\xi^p - \xi \in K$ since $\sigma(\xi^p - \xi) = (\xi+1)^p - (\xi+1)$

$= \xi^p - \xi + 1 - 1$

$= \xi^p - \xi$.

So $\xi \in \mathrm{Fix}(\sigma) = K$.

Thm 3.24 in GKS: let $\mathrm{char}(K) = p$. There is

a bijection:

$\{$1-dim subspaces $\langle c\rangle + \wp(K) \leqslant K/\wp(K)$ (over $\mathbb{F}_p$)$\} \leftrightarrow \{L/K$ Galois ext of deg $p\}$

$\langle c\rangle + \wp(K) \longmapsto K(a)$, where $\wp(a) = c$.

• **well defined**: if $c' \in (\langle c\rangle + \wp(K)) \setminus \wp(K)$, then

$c' = ac + \wp(b)$ with $b \in K$, $a \in \mathbb{F}_p^\times$.

Let α be an AS-root of c, set $\beta = a\alpha + b \in K(\alpha)$

we have $\alpha = a^{-1}(\beta - b)$, so $\alpha \in K(\beta)$, so $K(\alpha) = K(\beta)$.

$$\beta^p - \beta = a^p\alpha^p + b^p - a\alpha - b = \overset{\in \mathbb{F}_p!}{a}(\alpha^p - \alpha) + (b^p - b) = ac + \wp(\beta) = c'$$

injektiv:

• if $K(\alpha) = K(\beta)$, then $\beta = a\alpha + b$ with $a \in \mathbb{F}_p^\times$, $b \in K$

(why?) $\to$ AS27, Satz 2.

$\alpha^p - \alpha = c \in K$, $\beta^p - \beta = d \in K$:

$$d = \beta^p - \beta = a(\alpha^p - \alpha) + (b^p - b) = ac + \wp(b) \in \langle c\rangle + \wp(K)$$

So $\langle c\rangle + \wp(K) = \langle d\rangle + \wp(K)$.

• surjektiv: clear.

Lemma: (3.25 in GKS, Satz 3 in AS27).

Let K be a field of $\mathrm{char}(K) = p > 0$. If $\exists\ L/K$ Galois of deg p.

then $\exists\ M/L$ of deg p, M/K of deg p^2 cyclic.

Proof: by hand in AS27, by Hilbert 90 in GKS:

GKS 3.22: Hilbert 90: L/K finite cyclic, σ a generator of $\mathrm{Gal}(L/K)$, $a \in L$

$\mathrm{Tr}_{L/K}(a) = 0 \iff \exists\, b \in L\ \ a = \sigma(b) - b$ (where $\mathrm{Tr}_{L/K}(a) = \sum_{i=0}^{p-1} \sigma^i(a)$)

$L = K(\xi)$, $\xi^p - \xi = a \in k \implies -a^{-1} - a^{-1}\xi^{1-p} = -\xi^{-p} \iff \mathrm{Tr}(\xi^{-1}) = -a^{-1}$.

$\mathrm{Gal}(L/K) = \langle \sigma \rangle$

$$\implies -\xi^{p-1} = -1 - a\xi^{-1}$$

$$\mathrm{Tr}(\underbrace{-\xi^{p-1}}_{:= b}) = \mathrm{Tr}(-1 - a\xi^{-1}) = -\underbrace{\mathrm{Tr}(1)}_{=0} - a\,\underbrace{\mathrm{Tr}(\xi^{-1})}_{= -a^{-1}} = 1$$

$$\mathrm{Tr}(b^p - b) = \mathrm{Tr}(b^p) - \mathrm{Tr}(b) = 1 - 1 = 0 \overset{H90}{\implies} b^p - b = \sigma(c) - c$$

for some $c \in L$. $c \notin k$ bc $\sigma(c) \neq c$

<u>Claim</u>: let $y^p - y = c$, then $M = k(y)$ is as desired.

- c has no AS-root in L:

if $c = \wp(e)$, $e \in L$, then

$$\wp(b) = \cancel{\wp}\,\sigma(c) - c$$
$$= \sigma(\wp(e)) - \wp(e)$$

$\vdots$

$$\wp(b - \sigma(e) + e) = 0$$
$$b - \sigma(e) + e = h \in \mathbb{F}_p$$
$$\implies b - h = \sigma(e) - e$$
$$H90 \implies \mathrm{Tr}(b - h) = 0$$
$$\implies \mathrm{Tr}(b) = 0 \;\lightning$$

- $[M:K] = [M:L][L:K] = p^2$
- So, $\mathrm{Gal}(M/K) \cong \begin{cases} C_{p^2} \text{ or} \\ C_p \times C_p \end{cases}$

Assume $\mathrm{Gal}(M/K) = C_p \times C_p$, then Galois correspondence:

$$\begin{tikzcd}[column sep=small]
 & M \arrow[dl, dash, "p"'] \arrow[dr, dash, "p"] & \\
L' \arrow[dr, dash, "p"'] & \neq & L \arrow[dl, dash, "p"] \\
 & K &
\end{tikzcd}$$

$L' = \mathrm{Fix}(\{e\} \times C_p)$, $L = \mathrm{Fix}(C_p \times \{e\})$

So $L' = K(z)$, $z^p - z = d \in K$

& $L(z) = M$.

Now
$$\langle c \rangle + \wp(K) = \langle d \rangle + \wp(K)$$
since $L(y) = M = L(z)$

& $y^p - y = c$, $z^p - z = d$.

So $c = id + e^p - e$ for $i \in \mathbb{F}_p^\times$ & $e \in L$

(bc $c \in \langle d \rangle + \wp(L)$) & $b^p - b = \sigma(c) - c$

$$= \sigma(id + e^p - e) - id - e^p + e$$
$$= (\sigma(e) - e)^p - (\sigma(e) - e)$$
(bc $d \in K$).

So $\wp(b) = \wp(\sigma(e) - e)$

$\Rightarrow \wp(b - \sigma(e) + e) = 0$

$\Rightarrow b - \sigma(e) + e = j \in \mathbb{F}_p$.

$\Rightarrow \mathrm{Tr}(\sigma(e) - e) = 0$ (H90)

$\mathrm{Tr}(b + j) = \mathrm{Tr}(b) \neq 0$ ↯

Back to the thm: if Ω/P is finite (& not trivial), then

- P is perfect (easy)
- if $p \mid [\Omega : P]$, then $\mathrm{char}(P) \neq p$ (otherwise, I have p-cyclic ext.).

$\rightsquigarrow$ recovery infos about char from $\mathrm{Gal}(\Omega/P)$!!